When Does Leaf Geometry Help? Sweep-Valuation Attention for Studio-to-Field Medicinal Leaf Identification

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Abstract

Medicinal-leaf classifiers report near-perfect accuracy on studio images, but little is known about what they rely on or how they transfer to field photographs. We propose *sweep-valuation attention* (SVA), a backbone-agnostic module that sweeps a leaf along many directions and, for every slab of the sweep, measures the area and boundary length that enter a weighted cubical complex of the blade and of a learned venation field. Each measurement becomes a token that attends to the image patches of the same slab. We study SVA against matched controls (global fusion of the same measurements, slab attention without measurements, and strip pooling) on three backbones (ConvNeXt-T, ViT-S, BioCLIP 2), using a de-duplicated, leaf-grouped protocol for the 30-species Medicinal Leaf Dataset and transfer to 20 species of the DIMPSAR field photographs. Clean accuracy is saturated for all models. With ConvNeXt-T, SVA raises field-photo accuracy from 33.5% to 49.3% (15/15 runs) and keeps this gain without rotation augmentation, where it exceeds strip pooling and slab-only attention by 5–6 points. The gain disappears for ViT-S and for BioCLIP 2, which reaches 81% without any module. Ablations show that appearance-conditioned weights and a fine sweep drive the effect, whereas restricting attention to the slab adds little; Euler-characteristic tokens make models sensitive to holes in the blade but transfer worse. All code, run outputs, statistics and the paper sources are available at `https://github.com/Devguru-codes/research_paper_leaf`.

1 Introduction

Medicinal plants are identified in the field mostly from their leaves, and errors are not equally costly: confusing two culinary herbs is harmless, but confusing a remedy with a toxic look-alike, such as *Nerium oleander* with other Apocynaceae, can be dangerous [1]. Deep networks report 97.5–99.7% accuracy on Indian medicinal leaf benchmarks [2, 3]. Those numbers, however, are obtained on single leaves photographed on white backgrounds, typically with random image-level splits. They say little about what the models rely on or how they behave when the leaf is rotated, degraded or photographed in the field.

Botanists describe a leaf by its geometry: how wide the blade is along its length, where it tapers, how ragged its margin is, how its veins branch. These are measured *at particular places* and judged together with what the tissue looks like there. Standard CNN and ViT classifiers have no built-in notion of such object-centred, localised geometry. Topological and morphological summaries can supply it, and recent work fuses persistence diagrams or Euler characteristic transforms with deep features [4–7]. In these designs, however, the geometric signal enters as a global descriptor, so the place where geometry is measured is decoupled from the place where the visual model looks.

We propose *Sweep-Valuation Attention* (SVA), which couples the two. A directional sweep partitions the leaf into ordered slabs along K directions. For each slab, SVA measures how much area and boundary length of two weighted cubical complexes, the blade and a learned venation field, enter the filtration there. These quantities are the metric intrinsic volumes, which together

with the Euler characteristic span all motion-invariant planar valuations. The measurement forms the query of a sweep token, and the token's cross-attention to backbone patches is restricted by *the same* soft slab membership that defines the measurement. The resulting module (SV-Former) is differentiable, backbone-agnostic and costs one histogram and one matrix product per complex.

Because our questions are about mechanism, we design controls that hold everything else fixed: the same valuations fused globally (no co-registration), slab attention without valuations (geometry without measurement), and strip pooling [8] (localised directional pooling without an object-centred filtration). We also audit the data: we find exact duplicates and re-shots of the same leaf, and evaluate all models under a grouped protocol.

Contributions.

1. **SVA**, a module whose tokens carry filtration increments of the intrinsic volumes of appearance-weighted cubical complexes (blade and learned venation field) along a fine directional sweep, and attend within the corresponding filtration slab. To the best of our knowledge, we did not find prior work that explicitly co-registers filtration-derived valuation increments with attention restricted to the corresponding filtration slab.

2. **A controlled study** with matched controls on three backbones and five probes. With an ImageNet CNN, SVA improves studio-to-field transfer by 16 points and keeps this gain without rotation augmentation, where it beats strip pooling and slab-only attention by 5–6 points. We also report where it does not help: on ViT-S and BioCLIP 2 the gain vanishes, and the ablations show that appearance-conditioned weights and a fine sweep, not slab-restricted attention, drive the effect.

3. **A leakage-audited protocol and probes** for the Medicinal Leaf Dataset: duplicate and re-shot grouping, a corruption suite (MedLeaf-C), a leaf cue-conflict set and a topology-destruction test with perceptually matched edits.

Code, splits, probes and all run outputs are released.

2 Related work

Medicinal leaf identification. Work on Indian medicinal leaves has moved from hand-crafted shape, colour and texture descriptors to transfer learning. DeepHerb combined Xception features with SVMs [2]; later CNN, hybrid CNN–transformer and feature-fusion models report 98.9–99.7% on this and related datasets [3], typically with random image-level splits. Closest to our biological motivation, a dual-branch network fuses an RGB branch with a branch on skeletonised veins by late concatenation [9]. A cascaded taxonomic classifier targets open-world medicinal plants [10]. We reproduce DeepHerb-style and dual-branch pipelines under our leakage-audited protocol.

Euler characteristic transforms. The ECT and the persistent homology transform encode a shape through sublevel-set topology along all directions [11]. DECT makes the ECT differentiable by sigmoid relaxation and learns directions end-to-end [12]. WECT weights cells by image intensity [13]. Blaser et al. encode the ECT as per-direction token sequences, including convolutional and complex-valued variants that are equivariant to planar rotations [7]. For plants, Ayub et al. show that ECT images of *blade contours* classify complex leaf shapes and even predict venation from blade geometry [14]. Their ECT is computed on the outline only; venation is a prediction target, and appearance is not used. We build on DECT's relaxation and on these encodings. We do not claim weighted, tokenised or rotation-equivariant ECTs, or ECTs of leaves, as contributions.

Topology meets attention. Topological summaries are increasingly fused with deep features. PHG-Net injects persistence-diagram features into CNN/transformer classifiers [4]; TDA-ViT fuses topological and ViT features [5]; GraPHFormer aligns graph topology and vision representations [15]. Gu et al. process multi-filtration cubical persistence diagrams with cross-attention and fuse the embedding with CNN/transformer features [6]. TopoFormer slices graphs along filtrations into attention-friendly topological token sequences [16]. For time series, Faghihi and Saki add persistent-homology and directional Euler statistics of each input window as additive biases on attention logits [17]; the topology summarises the whole window rather than defining where each query may attend. In these methods the topological signal enters as a *global* descriptor or as tokens among themselves, so where the visual model looks is not tied to where the topology was measured. SVA differs in exactly this respect: each filtration slab yields both a valuation increment and the spatial support of that token's attention. Our controls (global fusion of the same valuations; slab attention without valuations) separate the two ingredients.

Intrinsic volumes and directional pooling. Area, boundary length and Euler characteristic (the planar intrinsic volumes, or Minkowski functionals) are classical morphological descriptors with efficient digital estimators [18, 19], and the perimeter of a planar shape can be recovered from its ECT [20]; by Hadwiger's theorem they span all continuous motion-invariant valuations [21, 22]. We use them as filtration increments along a sweep, not as global descriptors, and do not claim them as new features. Strip pooling [8] and axial attention aggregate features along image rows and columns. Our strip control instantiates this family; unlike SVA, its strips are fixed to the image axes, carry no measured geometry and are not defined by a filtration of the object.

Taxonomy and uncertainty. Hyperbolic and hierarchy-aware embeddings have been applied to plant and biological taxonomies [23, 24], and BioCLIP 2 learns hierarchical contrastive representations at scale [25]. Conformal prediction for long-tailed plant identification [26] and for hierarchical labels [27, 28] is established. Our deployment layer uses these ideas; its only addition is a hazard-asymmetric Mondrian calibration for toxic look-alikes. We treat it as an application component, not as a methodological contribution.

Shape versus texture. ImageNet CNNs are biased towards texture [29]. We adapt the cue-conflict paradigm to leaves (silhouette of one species filled with the lamina of another) and add a topology-destruction probe, in order to test what SV-Former actually relies on.

3 Data and leakage-audited protocol

Dataset. We use the segmented subset of the Medicinal Leaf Dataset of Roopashree and Anitha [30] (CC BY 4.0): 1,835 RGB photographs (1600×1200) of single leaves from 30 Indian medicinal species on a white background, with 34–122 images per species ($3.6\times$ imbalance). The leaf covers 4.4–68.5% of the frame (median 34.8%). The species span 28 genera, 19 families and 12 orders of the APG IV system, with genuine sibling groups (e.g. Apocynaceae: *Nerium*, *Tabernaemontana*, *Carissa*; Myrtaceae: *Psidium* and two *Syzygium*; Moraceae: two *Ficus* and *Artocarpus*; Lamiaceae: *Mentha*, *Ocimum*, *Plectranthus*); see table 3.

Audit. Reported accuracies on this dataset are near saturation (97.5–99.7% [2, 3]) and use random image-level splits. Our audit found three issues. (i) *Exact duplicates*: 10 pairs of identical

photographs saved under different names, all in *Carissa carandas* (1,825 images remain after removal). (ii) *Re-shots of the same physical leaf*: some leaves were photographed more than once, at times under different illumination or orientation. Hashing alone cannot separate these from merely similar leaves (e.g. lanceolate *Alpinia* silhouettes), so we link two images when their raw-frame 1024-bit dHash distance is ≤ 20, or when SIFT matching with RANSAC yields ≥ 40 geometrically consistent inliers whose similarity transform has scale in $[0.8, 1.25]$ and whose convex hull covers $\geq 25\%$ of the leaf. These criteria were set by visual inspection and reject hub matches caused by repetitive texture (e.g. *Mentha*). The resulting 22 multi-image groups (45 images, at most 3 per group) are kept intact across splits. Over-grouping is conservative: it constrains splits but cannot leak. (iii) *Filename collisions*: *Psidium guajava* and *Punica granatum* share the `PG-S-xxx` prefix (65 colliding names), so files must be indexed by class and name.

Protocol. All results use stratified, group-aware 5-fold cross-validation on the 1,825 de-duplicated images. Inside each training part, one group-aware stratified holdout ($\approx 1/6$) serves as the calibration set for conformal prediction and for monitoring only. Epoch count and hyper-parameters are fixed in advance, with no model selection on test or calibration data. With random image-level splits, 2–4% of test images per fold have a duplicate or re-shot in the training part (section 6).

Preprocessing. The leaf mask is obtained by thresholding the white background (all channels < 235) followed by morphological cleaning. Each image is cropped to a square centred on the leaf whose inscribed circle contains the whole leaf, so that any rotation about the centre is lossless, and resized to 448^2 (training input 384^2). Vein pseudo-labels are multiscale Sato ridge responses on CLAHE-normalised luminance inside the eroded mask.

Cross-domain test set. For transfer to field photographs we use DIMPSAR [31] (CC BY 4.0), whose single-leaf subset was captured with smartphones against varied backgrounds (tiles, soil, cloth, paper). Vernacular folder names were mapped to species and checked visually. Twenty species overlap ours; note that Kannada *Arali* denotes *Ficus religiosa* (peepal), not *Nerium oleander*. Leaves are localised with our own masker, a ConvNeXt-T FPN trained only on our leaves composited onto DTD textures [32] and procedural backgrounds; no third-party segmentation model is used.

4 Method: Sweep-Valuation Attention

Figure 1 summarises SV-Former. An appearance backbone (CNN or ViT) produces patch tokens. Two structural fields are derived from the same image: the *blade*, a deterministic leaf mask, and a *learned venation field*, a vein-probability map predicted by a light head on early backbone features. A directional sweep partitions the leaf into ordered slabs. For every direction–slab pair, SVA measures how much of each field's *intrinsic volumes* (area and boundary length) enters the filtration inside that slab, and uses *the same slab* to restrict where the resulting token attends among the patch tokens.

4.1 Weighted cubical complexes

The leaf crop is resampled to a $G \times G$ grid (G=96). Pixel centres x_p are expressed in a frame centred at the blade centroid and scaled by its root-mean-square radius, which makes all quantities below invariant to translation and isotropic scale. We use the cubical complex whose 0-cells are

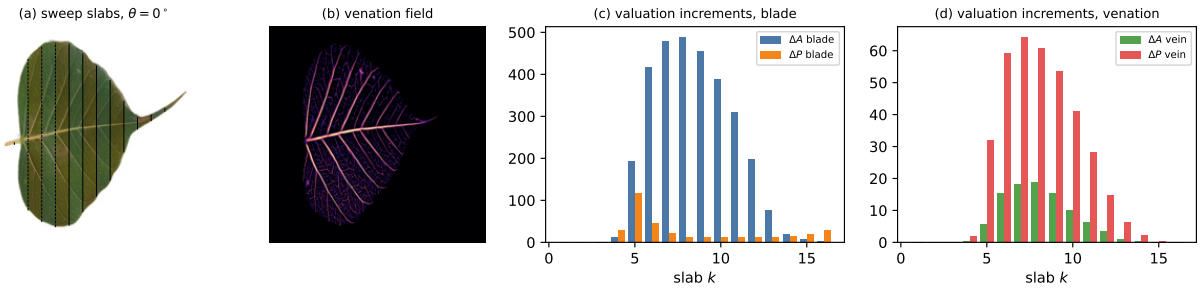


Figure 1: The sweep-valuation signal for one leaf (*Ficus religiosa*) and one of the K=32 directions. (a) The leaf is partitioned into T=16 slabs along θ in its centroid/RMS frame. (b) Ridge-filter vein pseudo-label that supervises the learned venation field. (c,d) Area and perimeter increments per slab for the blade and the venation field, shown here with unit weights; in SVA the cell weights are appearance-conditioned. Each slab's increments form the query of one token, which attends to the image patches of the same slab. The long perimeter in the last blade slabs comes from the drip tip.

pixels, whose 1-cells are pairs of 4-adjacent pixels and whose 2-cells are 2×2 pixel squares. Given per-pixel weights $a(p) \in [0, 1]$, cells inherit the minimum weight of their vertices,

$$w_0(p) = a(p), \qquad w_1(e) = \min_{v \in e} a(v), \qquad w_2(q) = \min_{v \in q} a(v). \tag{1}$$

This keeps face monotonicity $w(\text{coface}) \leq w(\text{face})$, so the weighted complex is a fuzzy subcomplex. For binary a it reduces exactly to the standard cubical complex of the set.

Appearance-conditioned weights. Two weight fields are built from mid-level backbone features f (bilinearly sampled at the grid), a small per-pixel MLP g and a sigmoid σ:

$$a_{\mathrm{b}}(p) = m(p)\, \sigma(g_{\mathrm{b}}(f(p))), \qquad a_{\mathrm{v}}(p) = m(p)\, v(p)\, \sigma(g_{\mathrm{v}}(f(p))), \tag{2}$$

where m is the blade mask and v is the predicted vein probability. The last layer of g is initialised so that $a \approx 0.88\, m$ at the start of training. The vein head is pseudo-supervised by a multiscale Sato ridge response [33] inside the eroded mask and also receives task gradients through SVA. In contrast to fixed intensity weights (WECT [13]), the weights are learned and conditioned on appearance. We stress that a_{v} is a *learned* venation field, not an observed vascular structure.

4.2 Valuations of a weighted cubical complex

Let $H_d = \sum_{c \in C_d} w_d(c)$ be the weighted number of d-cells. For a binary set the three quantities

$$A = H_2, \qquad P = 2H_1 - 4H_2, \qquad \chi = H_0 - H_1 + H_2 \tag{3}$$

are the area (number of unit squares), the boundary length in pixel units (every square contributes four edges, and an edge shared by two squares is interior) and the Euler characteristic. They are the cubical counterparts of the three planar intrinsic volumes. By Hadwiger's theorem [21, 22], every continuous, rigid-motion-invariant valuation on planar convex bodies is a linear combination of these three. Because (3) is linear in the cell weights, it extends directly to the fuzzy complex (1). Our proposed token uses area and perimeter, i.e. the two *metric* intrinsic volumes. The purely topological one, χ, is retained as an ablation (section 6.5), because in our pilots it did not add information beyond area and perimeter (section 6).

4.3 Filtration increments along a sweep

For a direction $\theta \in S^1$ we use the lower-star filtration, $h_\theta(v) = \langle x_v, \theta \rangle$ and $h_\theta(c) = \max_{v \in c} h_\theta(v)$. Following the sigmoid relaxation of DECT [12], a cell enters the filtration at threshold t with soft indicator $\sigma(\lambda(t - h_\theta(c)))$. Fix thresholds $t_0 < t_1 < \cdots < t_T$, where $t_1, \ldots, t_{T-1}$ are uniform on $(-R, R)$, $t_0 = -\infty$ and $t_T = +\infty$. The soft membership of height h in slab k is

$$s_k(h) = \sigma\big(\lambda(t_k - h)\big) - \sigma\big(\lambda(t_{k-1} - h)\big), \qquad k = 1, \ldots, T, \tag{4}$$

and because the outer thresholds are infinite, $\sum_k s_k(h) = 1$ for every h. The *filtration increment* of the weighted d-cell count in slab k is $\Delta H_d(\theta, k) = \sum_{c \in C_d} w_d(c)\, s_k(h_\theta(c))$, and the valuation increments follow by linearity:

$$\Delta A(\theta, k) = \Delta H_2, \qquad \Delta P(\theta, k) = 2\Delta H_1 - 4\Delta H_2, \qquad \Delta\chi(\theta, k) = \Delta H_0 - \Delta H_1 + \Delta H_2. \tag{5}$$

Each is the change of the corresponding relaxed valuation of the sublevel set between t_{k-1} and t_k, and $\sum_k$ telescopes to the total valuation of the leaf, independently of θ. We compute (5) for the blade (a_b) and the venation field (a_v) with K uniform directions and T slabs (K=32, T=16, R=2, $\lambda = \kappa T / 2R$ with κ=4).

Sweep-valuation token. The structural feature of token (θ, k) is the 8-vector of area and perimeter increments of both fields together with their cumulative values up to slab k,

$$\phi(\theta, k) = \mathrm{slog}\Big[\Delta A_\mathrm{b}, \Delta P_\mathrm{b}, \Delta A_\mathrm{v}, \Delta P_\mathrm{v},\ \textstyle\sum_{j \le k}(\cdot)\Big], \qquad \mathrm{slog}(x) = \mathrm{sign}(x)\log(1 + |x|). \tag{6}$$

The increments describe the local width profile and boundary complexity of the slab; the cumulative values locate it within the leaf. The Euler variant replaces ϕ by $[\Delta\chi_\mathrm{b}, \Delta\chi_\mathrm{v}, \chi_\mathrm{b}(t_k), \chi_\mathrm{v}(t_k)]$, and the Hadwiger variant uses all three valuations (12 features).

Efficient computation. Directions are fixed and coordinates come from the mask, so the heights carry no gradient; only the weights do. We hard-bin heights into n_b=256 fine bins, scatter-add cell weights per dimension into a per-direction histogram, and apply the slab kernel (4) evaluated at bin centres with one matrix product (the kernel is built in double precision). This costs $O(BKN)$ memory for N cells, needs no sorting, and is exact in the telescoping and sharp limits (verified by unit tests, section A).

4.4 Co-registered slab attention

Each direction–slab pair (θ, k) defines one sweep token with query $q_{\theta k} = \mathrm{MLP}(\phi(\theta, k)) + e_k$, where e_k is a learned slab-index embedding. The token cross-attends to the backbone patch tokens $\{z_p\}$ with an additive bias derived from the *same* soft slab membership (4), evaluated at patch centres:

$$\mathrm{attn}_{(\theta,k)}(p) \propto \exp\Big(\tfrac{\langle W_Q q_{\theta k}, W_K z_p \rangle}{\sqrt{d}} + \log\big(s_k(\langle x_p, \theta \rangle)\, \bar{m}(p) + \epsilon\big)\Big), \tag{7}$$

where $\bar{m}(p)$ is the fraction of leaf pixels in patch p and $\epsilon = 10^{-4}$. The structural measurement and the visual evidence attached to it are thus taken over the same filtration-defined region. To the best of our knowledge, we did not find prior work that explicitly co-registers filtration-derived valuation increments with attention restricted to the corresponding filtration slab (section 2). The attention output is added to the query with a mass gate $M/(M + 0.5)$, $M = \sum_p s_k \bar{m}$, so that slabs outside the leaf contribute little. The KT=512 tokens are mixed by a two-layer *sweep transformer* whose learned relative position bias depends only on $(\Delta\theta \bmod K, \Delta k)$. Tokens are mean-pooled and concatenated with the backbone's global token for classification. Training minimises cross-entropy (label smoothing 0.1) plus 0.2× the masked BCE of the vein head.

4.5 Symmetry properties

Proposition 1. *Let $\mathcal{R}$ be a rotation by a multiple of 90° acting on the pixel grid and let the weights be unit (binary blade). Then ΔA, ΔP and $\Delta\chi$ of $\mathcal{R}I$ equal those of I cyclically shifted by $K/4$ along θ for each multiple of 90°; for rotations by $2\pi/K$ the same holds up to grid resampling.*

The proof follows from $\langle \mathcal{R}x, \theta\rangle = \langle x, \mathcal{R}^{-1}\theta\rangle$, the equivariance of the centroid/RMS frame, and the fact that a 90° grid rotation maps cells of each dimension bijectively onto cells of the same dimension (so the counts H_d in every slab are permuted, not changed). It is checked numerically in section A. The appearance-conditioned weighted signal and the full SV-Former inherit the rotation behaviour of the backbone ($f(\mathcal{R}I) \neq \mathcal{R}f(I)$ in general). We therefore do *not* claim exact invariance for them and evaluate their behaviour empirically (section 6). Rotation-equivariant ECT tokenisation as such is not our contribution [7].

4.6 Controls built from the same code

To isolate what matters, we compare five models that share the backbone, vein head, training recipe and, where applicable, the token width and transformer depth:

1. **plain**: backbone global token only;
2. **global**: the full $K \times T \times 8$ valuation grid flattened through an MLP and fused globally (in the spirit of [4, 6]); same structural content, no co-registration;
3. **strip**: strip pooling [8], i.e. mask-weighted means of patch tokens over every row and column of the patch grid, with learned positions and the same two-layer transformer; localised directional pooling without an object-centred sweep or structural content;
4. **slab-only**: identical to SVA except that $\phi \equiv 0$, so each query is only the slab embedding e_k; the same co-registered slab geometry, no valuation content;
5. **SVA** (proposed).

Model 2 versus 5 asks whether co-registration matters; 3 versus 5 asks whether an object-centred filtration sweep matters beyond generic directional pooling; 4 versus 5 asks whether the measured valuations matter beyond the slab geometry itself.

5 Experimental setup

Backbones and training. The main backbones are ConvNeXt-T (ImageNet-1k), ViT-S/16 (AugReg, ImageNet-21k→1k) and the BioCLIP 2 ViT-L/14 visual tower (all but the last two blocks frozen). All variants of a backbone share the recipe: AdamW (new modules lr 10^{-3}, weight decay 0.05), 2 warm-up epochs then cosine decay, batch 32, mixed precision, label smoothing 0.1. The backbone lr is 10^{-4} for ConvNeXt-T and 5×10^{-5} for ViT-S, both trained 40 epochs at 384^2; BioCLIP 2 uses lr 3×10^{-5}, 20 epochs and 224^2 input. Inputs are converted to each backbone's own normalisation statistics. Augmentation uses uniform rotation in $[-180^\circ, 180^\circ]$, scale $[0.85, 1.15]$, translation $\pm 5\%$, horizontal flips and mild colour jitter. Geometric transforms are applied jointly to image, mask and vein pseudo-label. No early stopping or test-time augmentation is used. Runs use Kaggle T4 GPUs.

Metrics and statistics. We report accuracy, balanced accuracy, macro-F1, expected calibration error (15 bins) and *mistake severity*, i.e. the mean taxonomic distance (1 = same genus … 4 = different order) between predicted and true species over errors. All splits are grouped stratified 5-fold cross-validation over the deduplicated, leaf-grouped data (section 3). For ConvNeXt-T the five models are trained with 3 seeds per fold (15 runs each); the full probe suite is run on seed 0 and the cross-domain test on all seeds. ViT-S and BioCLIP 2 use one seed per fold. Ablations use ConvNeXt-T, seed 0, folds 0–2. Paired comparisons use the Wilcoxon signed-rank test over fold–seed pairs and McNemar's test on pooled predictions.

Ablations. Starting from SVA: Euler-only and full-Hadwiger tokens; blade-only and vein-only fields; unit cell weights; *permuted* learned weights (the same weight values shuffled within the leaf before computing the valuations, which keeps their distribution but destroys the weight–location correspondence); global instead of slab-restricted attention; a coarser sweep (K=16, T=8); and a softer or sharper relaxation ($\kappa \in \{1, 16\}$). A separate arm trains plain, strip, slab-only and SVA on all folds and three seeds *without* rotation augmentation, to measure robustness that is built in rather than learned from data.

Mechanistic probes. All probes are applied to each fold's test images with fixed seeds.

- **Rotation sweep**: accuracy at $0°, 30°, \ldots, 330°$ and the fraction of images whose prediction is identical at all 12 angles.
- **MedLeaf-C**: ten corruptions × three severities: Gaussian noise, blur, JPEG, low resolution, darkening, contrast loss, hue shift (yellowing), desaturation (severity 3 is grayscale), occluders, and pasted procedural/soil-like backgrounds. Masks are the oracle masks transformed consistently.
- **Cue conflict**: the silhouette of a test leaf A is filled with the lamina of a test leaf B of another species. B is aligned by principal axes and enlarged until it covers A, so that B's outline contributes no shape cue. Shape bias $= n_A/(n_A + n_B)$ where n_A (n_B) counts predictions of A's (B's) species. We also report it on pairs with anisotropic stretch ≤ 2.
- **Topology destruction**: appearance-preserving edits that change topology versus matched controls that do not. For the blade: three interior holes (exact $\Delta\chi = -3$, verified) versus three boundary bites of equal removed area ($\Delta\chi = 0$, verified). For venation: inpainting eight small disks on strong veins versus eight on vein-free lamina. Every edit records LPIPS, pixel RMSE and a colour-histogram distance. The response (KL divergence between clean and edited predictions) is analysed by regression on visual distance with a topology-changing indicator.
- **Cross-domain**: models trained on the segmented images are tested on the 20 overlapping DIMPSAR species, both on auto-masked crops with the background whitened and on raw crops with the background kept (30-way and 20-way-restricted accuracy).

6 Results

All numbers below are generated from the run outputs by the released scripts; ± is the standard deviation over runs. Paired tests compare SVA with each control on identical folds and seeds (Wilcoxon signed-rank; with five folds the smallest attainable two-sided p is 0.0625, so for five-fold comparisons we also report win counts).

Table 1: Main results (%). Clean = in-domain grouped 5-fold accuracy; Rot. min = worst accuracy over 12 rotations; MedLeaf-C = mean over 10 corruptions × 3 severities; Shape bias from the cue-conflict set; Field = 20-way accuracy on DIMPSAR field photographs (auto-masked or raw). ConvNeXt-T: Clean and Field over 5 folds × 3 seeds, other columns seed 0; ViT-S and BioCLIP 2: 5 folds, seed 0.

Backbone	Model	Clean	Rot. min	MedLeaf-C	Shape bias	Field (mask)	Field (raw)
ConvNeXt-T	Plain	99.9 ± 0.1	99.9 ± 0.2	81.1 ± 1.1	6.5 ± 1.5	35.1 ± 1.3	33.5 ± 1.2
	Global fusion	100.0 ± 0.1	99.7 ± 0.3	81.0 ± 0.8	7.0 ± 1.9	34.9 ± 1.7	33.7 ± 1.3
	Strip pooling	100.0 ± 0.0	99.9 ± 0.1	87.6 ± 0.8	9.1 ± 1.5	42.9 ± 1.6	47.8 ± 2.0
	Slab-only	100.0 ± 0.1	100.0 ± 0.0	89.4 ± 0.9	7.4 ± 1.9	42.2 ± 1.1	45.7 ± 2.0
	SVA (ours)	100.0 ± 0.1	99.9 ± 0.1	88.6 ± 0.5	9.0 ± 2.3	43.1 ± 2.8	49.3 ± 2.8
ViT-S/16	Plain	99.9 ± 0.1	99.9 ± 0.1	96.4 ± 0.3	9.7 ± 1.0	51.6 ± 0.4	60.4 ± 0.9
	Global fusion	100.0 ± 0.0	99.9 ± 0.1	96.2 ± 0.6	9.1 ± 1.3	50.1 ± 0.7	58.6 ± 1.8
	Strip pooling	99.9 ± 0.1	99.9 ± 0.1	96.2 ± 0.5	10.3 ± 1.3	51.2 ± 1.1	62.7 ± 1.0
	Slab-only	100.0 ± 0.0	99.8 ± 0.1	96.6 ± 0.5	11.0 ± 1.4	52.0 ± 0.8	64.8 ± 2.1
	SVA (ours)	99.9 ± 0.1	99.9 ± 0.1	96.6 ± 0.3	9.3 ± 1.8	51.9 ± 1.0	60.7 ± 1.4
BioCLIP 2	Plain	99.9 ± 0.1	99.0 ± 0.4	93.8 ± 0.5	14.0 ± 1.3	58.9 ± 0.5	81.4 ± 0.7
	Global fusion	99.9 ± 0.1	99.0 ± 0.4	93.8 ± 0.6	14.3 ± 2.3	59.7 ± 1.1	81.3 ± 0.2
	Strip pooling	99.9 ± 0.1	99.2 ± 0.5	93.8 ± 0.2	14.2 ± 1.7	58.9 ± 0.7	81.5 ± 0.5
	Slab-only	100.0 ± 0.0	98.8 ± 0.5	93.9 ± 0.6	13.9 ± 1.2	60.7 ± 0.7	81.8 ± 1.4
	SVA (ours)	100.0 ± 0.0	98.5 ± 0.5	93.5 ± 0.5	13.7 ± 2.0	59.8 ± 1.1	81.7 ± 0.8

6.1 In-domain accuracy is saturated

On the de-duplicated, grouped protocol every model on every backbone reaches 99.9–100% clean accuracy and $\geq 98.5\%$ accuracy under any rotation (table 1). The benchmark therefore cannot separate architectures on clean accuracy, which motivates the robustness and transfer probes below. None of the structural modules costs clean accuracy.

6.2 Transfer to field photographs

Figure 2 shows the main effect. With ConvNeXt-T, which is trained on studio images only, SVA raises 20-way accuracy on raw field photographs from 33.5% (plain) to 49.3% (+15.8 points, 15/15 fold–seed pairs, $p < 10^{-3}$) and on auto-masked photographs from 35.1% to 43.1% (+8.0, 15/15, $p < 10^{-4}$). Fusing the same valuations globally gives no gain (33.7% raw), so the improvement is not explained by the presence of shape features alone. Against the localised controls the picture is more nuanced. SVA beats slab-only attention by +3.6 points on raw photographs (12/15, $p = 0.003$), but its lead over strip pooling (+1.5, 10/15, $p = 0.15$) is not significant when all models are trained with rotation augmentation.

The gains do not carry over to stronger backbones. With ViT-S, slab-only attention is best on raw photographs (64.8% vs. 60.4% plain), and SVA is level with plain (60.7%). BioCLIP 2 reaches 81.4% on raw photographs without any module, about 30 points above the best ImageNet ConvNeXt model, and all five variants lie within 0.5 points of each other. A biology foundation model thus closes most of the studio-to-field gap on its own, and structural pooling adds nothing measurable on top of it.

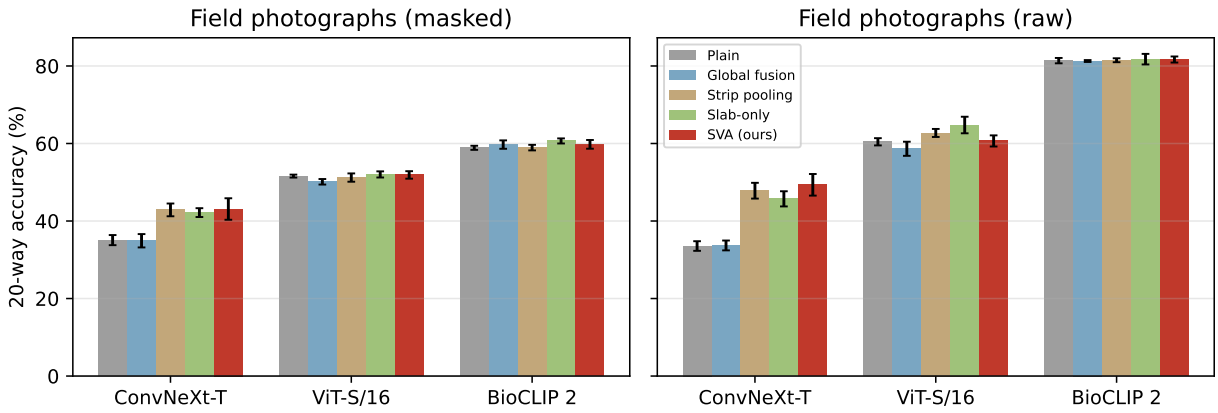


Figure 2: Studio-to-field transfer: 20-way accuracy on DIMPSAR photographs with the background whitened by our masker (left) or kept (right). Error bars: std over runs.

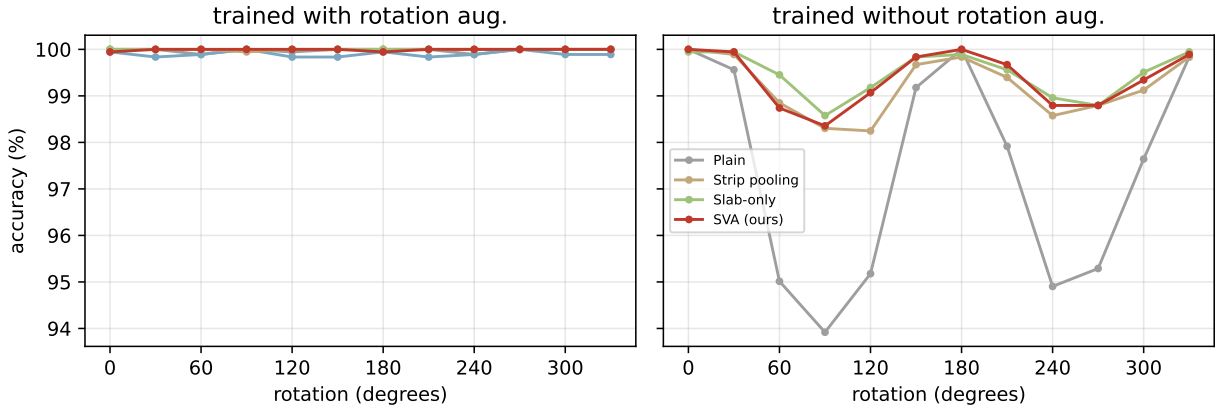


Figure 3: Accuracy versus test-time rotation (ConvNeXt-T, seed 0, mean over 5 folds), trained with (left) and without (right) rotation augmentation.

6.3 Learning without rotation augmentation

To test which robustness is built in rather than learned from augmented data, we retrain plain, strip, slab-only and SVA ConvNeXt-T models without rotation augmentation (fig. 3). All three localised modules remain far more rotation-robust than the plain model (worst-case accuracy 97.6–98.3% vs. 93.8%), and SVA is not more rotation-robust than slab-only attention (98.3% each). On field photographs, however, SVA keeps almost all of its transfer gain without augmentation (48.6% raw vs. 49.3% with augmentation), whereas strip pooling drops from 47.8% to 42.7%. Without augmentation SVA exceeds strip pooling by +5.8 (raw) and +4.0 (masked) points and slab-only attention by +5.2 (raw) points, winning 5/5 folds in each case. The object-centred sweep with measured valuations thus provides, by construction, transfer that image-axis strips only acquire through augmentation. [Seeds 1–2 of the no-augmentation arm pending.]

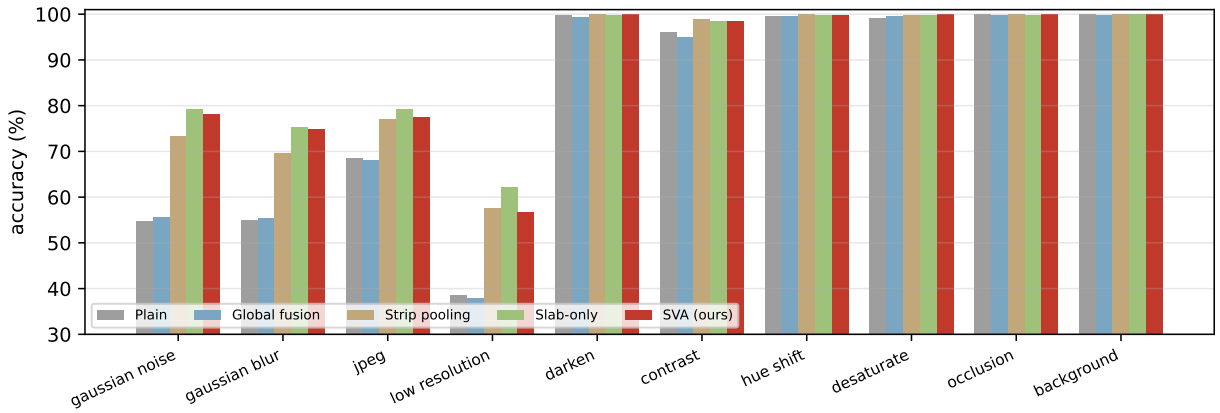


Figure 4: MedLeaf-C accuracy per corruption family (ConvNeXt-T, severity-averaged).

6.4 Corruptions and cue conflict

On MedLeaf-C, both slab-based models and strip pooling improve ConvNeXt-T by 6.5–8.3 points over plain and global fusion (fig. 4); slab-only attention is best (89.4%) and SVA is within one point (88.6%). The gains come from noise, blur, JPEG and low-resolution corruptions, i.e. from degraded texture; colour, occlusion and background corruptions are near ceiling for all models. For ViT-S and BioCLIP 2 all variants are within 0.5 points. Shape bias remains low everywhere (7–14%): when a silhouette is filled with another species' lamina, all models mostly follow the lamina. SVA raises ConvNeXt-T shape bias from 6.5% to 9.0% (5/5 folds), the same as strip pooling, and BioCLIP 2 is the most shape-biased backbone (14%). We therefore do not claim that SVA makes models shape-based; it makes ImageNet CNN features less brittle under texture degradation and domain shift.

6.5 What drives the gain: ablations

Table 2 varies one ingredient at a time (ConvNeXt-T, seed 0, folds 0–2; differences below about 3 points are within seed noise, cf. the seed std in table 1). Three ingredients matter. (i) *Learned, appearance-conditioned cell weights*: with unit weights, i.e. pure mask geometry, raw field accuracy falls from 53.1% to 43.5%; permuting the learned weights within the leaf costs 2.3 points, so their location carries part of the signal. (ii) *A fine sweep*: halving directions and slabs costs 6.3 points. (iii) *Both structural fields*: blade-only and vein-only variants lose 4.7 and 2.4 points. By contrast, restricting attention to the slab is *not* essential once the tokens carry valuations: attention over all leaf patches loses only 1.1 points. The co-registered attention contributes mainly in the slab-only model, where the slab is the only structural information. Finally, the choice of valuation matters. Euler tokens and the full Hadwiger token transfer worse than area and perimeter alone (45.7% and 44.3% raw).

6.6 Topology sensitivity

The topology-destruction probe separates what the models respond to. Punching three interior holes ($\Delta\chi = -3$) changes predictions less than removing the same area as three bites from the margin ($\Delta\chi = 0$) for every ImageNet model; in a regression of the response on visual distance and a topology-change indicator the indicator is negative (ConvNeXt-T: -0.5 to -1.4 log-units). SVA

Table 2: Ablations of SVA (ConvNeXt-T, seed 0, folds 0–2, %), sorted by raw field accuracy.

Variant	Clean	Rot. min	Field (mask)	Field (raw)
SVA (full)	99.9 ± 0.2	99.9 ± 0.2	47.0 ± 0.7	53.1 ± 2.7
softer relaxation (κ=1)	100.0 ± 0.0	99.9 ± 0.2	46.8 ± 1.1	52.5 ± 2.1
attention over all leaf patches	100.0 ± 0.0	100.0 ± 0.0	47.1 ± 1.1	52.0 ± 2.5
weights permuted within the leaf	100.0 ± 0.0	100.0 ± 0.0	44.5 ± 0.8	50.8 ± 1.1
venation field only	99.9 ± 0.2	99.8 ± 0.2	45.3 ± 0.6	50.7 ± 2.5
sharper relaxation (κ=16)	100.0 ± 0.0	99.9 ± 0.2	46.5 ± 2.6	50.6 ± 3.8
blade only	100.0 ± 0.0	99.9 ± 0.2	43.8 ± 2.0	48.4 ± 4.1
coarse sweep (K=16, T=8)	100.0 ± 0.0	99.9 ± 0.2	43.2 ± 1.2	46.8 ± 1.0
Euler token	100.0 ± 0.0	100.0 ± 0.0	43.0 ± 0.7	45.7 ± 2.8
Hadwiger token (area, perimeter, Euler)	100.0 ± 0.0	100.0 ± 0.0	38.9 ± 1.1	44.3 ± 1.5
unit cell weights	99.9 ± 0.2	99.8 ± 0.2	41.5 ± 1.2	43.5 ± 0.7

reacts most to margin bites (mean KL 35×10^{-4} vs. 17–24 $\times 10^{-4}$ for the controls), consistent with tokens that measure boundary length. The only model that is more sensitive to holes than to bites is the Euler-token variant (KL 46×10^{-4} for holes, 4.5 times any other model). The probe therefore confirms that the measured valuation shapes what the model is sensitive to, and it shows that Euler sensitivity, while real, does not help species transfer. Severing veins changes all models' predictions more than equal-size edits of vein-free lamina, by a similar factor for every variant, so venation sensitivity is a property of the backbones rather than of the module.

6.7 Leakage audit

[Leakage-audit numbers (DeepHerb / dual-branch reproductions, random vs. grouped) pending.]

7 Discussion

When does structure help? The clearest pattern in our results is a dependence on the backbone. For an ImageNet CNN, measuring area and boundary length along an object-centred sweep, with cell weights conditioned on appearance, turns a model that transfers poorly from studio to field photographs (34%) into one that transfers substantially better (49%), and it does so without needing rotation augmentation. For ViT-S the module is neutral, and for BioCLIP 2, whose pre-training already covers millions of organism images, no structural module changes anything. Practitioners with access to a biology foundation model should use it; the structural module is useful where such a model is unavailable or too large, e.g. for on-device CNNs.

What the ablations say about the mechanism. Our initial hypothesis was that *co-registration*, i.e. attending only within the slab whose geometry is measured, would be the key ingredient. The ablations do not support this as the main driver: once the tokens carry valuations, global attention performs within noise of slab-restricted attention. What matters is (i) appearance-conditioned weights, which let the network decide which parts of the blade and venation count towards the measured geometry, and (ii) a fine sweep. Restricting attention to the slab matters when the slab is the *only* structural information (slab-only model). We report co-registration as a design choice of SVA and not as the source of its gains.

Topology is measurable but not what transfers. Euler-characteristic tokens make a model demonstrably sensitive to holes in the blade, which none of the other models are, yet they transfer worse to field photographs than metric valuations. For species identification the metric profile of a leaf (how its width and margin length vary along each direction) is more informative and more robust than its connectivity. This is consistent with botanical practice, where outline and margin are primary characters, and with recent findings that leaf outlines are highly informative of other leaf traits [14].

Benchmark saturation. Clean accuracy on this dataset is saturated even after removing duplicates and grouping re-shots of the same leaf, and so are rotation robustness under rotation augmentation and several corruption families. Reports of 98–100% on studio leaf benchmarks should therefore be read as a lower bar; transfer to field photographs remains far from solved (49% for the best ImageNet model and 82% for BioCLIP 2 on 20 species).

Limitations. (i) One studio dataset and one field dataset with 20 shared species; field labels come from vernacular folder names that we mapped and checked visually. (ii) ViT-S and BioCLIP 2 were run with one seed, ablations with one seed on three folds; effects below about 3 points in those settings are within seed noise. (iii) The venation field is learned from ridge-filter pseudo-labels and is not a validated vein segmentation. (iv) Hazard tiers are an explicit assumption table; the conformal layer (section B) is limited by the small number of calibration images for toxic species and gives no guarantee under shift.

8 Conclusion

We introduced sweep-valuation attention, which turns a directional filtration of a leaf into tokens that carry the area and boundary length entering each slab, weighted by appearance, and attend to the matching image region. In a controlled study with matched controls on three backbones, it gives an ImageNet CNN a large and statistically robust gain in studio-to-field transfer (+16 points) and keeps that gain without rotation augmentation, where strip pooling does not. The same study shows the limits: the gain vanishes for a ViT and for the BioCLIP 2 foundation model, co-registered attention is not its main source, and Euler-characteristic tokens, although they make the model sensitive to topology, transfer worse than metric valuations. We release the leakage-audited protocol, the corruption, cue-conflict and topology-destruction probes, and all code and run outputs.

Code and data availability

All code, configurations, unit tests, the outputs of every training run (metrics, probe results, per-image predictions and logs), the statistical analyses and the LaTeX sources of this paper are publicly available at `https://github.com/Devguru-codes/research_paper_leaf`. Every number reported here is regenerated from those run outputs by the scripts in the repository, and the properties stated in section 4 are verified by its unit tests (section A). The image datasets are not redistributed; they are available from their original sources (section 3) [30, 31].

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A Unit tests of the structural signals

All properties used in section 4 are checked by automated tests that ship with the code (32 tests, all passing).

- **Exact Euler characteristic.** On toy shapes (disk, annulus, two disks, a lens-shaped leaf, and the lens with two holes; $\chi = 1, 0, 2, 1, -1$) the cubical $\chi = V - E + F$ of the binary set equals the value computed independently with scikit-image (4-connected foreground).
- **Telescoping.** For every direction, $\sum_k \Delta\chi(\theta, k)$, $\sum_k \Delta A(\theta, k)$ and $\sum_k \Delta P(\theta, k)$ equal the total valuation of the shape, to numerical precision.
- **Partition of unity.** Slab memberships (4) sum to one for every height.
- **Sharp limit.** With binary weights and large κ, the relaxed increments equal those of a hard sweep that assigns each cell to the slab containing its height.
- **Closed form.** For an $n \times n$ block of pixels, the totals are $(A, P, \chi) = ((n-1)^2, 4(n-1), 1)$.
- **Consistency.** The Euler column of the Hadwiger increments equals the separately implemented Euler increments.
- **Rotation.** For unit weights, rotating the input by 90° cyclically shifts $\Delta\chi$, ΔA and ΔP by $K/4$ directions (proposition 1).
- **Fuzzy subcomplex.** Learned cell weights satisfy $w(\text{coface}) \leq w(\text{face})$, and gradients reach the per-pixel weights.
- **Splits.** Under the grouped protocol no exact duplicate or re-shot group crosses a fold boundary and every image is tested exactly once; under the random protocol leaks occur.
- **Conformal.** On synthetic exchangeable data, mean marginal coverage over 30 repetitions is at least $1 - \alpha$ (up to 0.5 points). When a hazard tier's calibration set is too small for its target level α_τ, its threshold is infinite and all of its species are always included in the set, i.e. the procedure fails safe.

B Hazard-aware conformal sets

We calibrate split-conformal APS sets [34] (cf. hierarchical variants [27, 28]) on each fold's held-out calibration split (245 images), Mondrian over hazard tiers, with $\alpha = 0.10$ for all tiers (standard) or $\alpha = 0.10/0.05/0.01$ for tiers 0/1/2 (hazard-aware). Under exchangeability this gives tier-wise

coverage $\geq 1 - \alpha_\tau$; no guarantee is claimed under distribution shift. Two limitations dominate. (i) In-domain accuracy is saturated, so in-domain coverage is trivially 100% (mean set size 1.6–2.7 species). (ii) Each calibration split contains only about ten *Nerium* images; with $\alpha_2 = 0.01$ the tier-2 threshold is infinite, so oleander is included in every set, and even with $\alpha = 0.10$ it appears in 61% of in-domain sets because label smoothing keeps top-class probabilities near 0.9. On field photographs (no guarantee), standard sets cover 86–92% of true species for ConvNeXt-T and 95–97% for BioCLIP 2 (raw photos), but they are large (8–17 of 30 species, rarely resolving below order level). The sets therefore signal the shift honestly, but a deployable hazard-aware layer needs more calibration data for toxic species than this dataset provides. Full results are in `results/conformal.csv`.

C Species, taxonomy and hazard tiers

Table 3: The 30 species (APG IV order and family), images after de-duplication, and the hazard tiers assumed by the conformal layer (sources in the released code).

Order / family	Species	Name	n	Hazard
Brassicales / Brassicaceae	*Brassica juncea*	Indian Mustard	34	
Brassicales / Moringaceae	*Moringa oleifera*	Drumstick	77	
Caryophyllales / Amaranthaceae	*Amaranthus viridis*	Arive-Dantu	122	
Caryophyllales / Basellaceae	*Basella alba*	Basale	103	
Fabales / Fabaceae	*Pongamia pinnata*	Indian Beech	61	caution
Fabales / Fabaceae	*Trigonella foenum-graecum*	Fenugreek	36	
Gentianales / Apocynaceae	*Carissa carandas*	Karanda	64	
Gentianales / Apocynaceae	*Nerium oleander*	Oleander	62	high
Gentianales / Apocynaceae	*Tabernaemontana divaricata*	Crape Jasmine	56	caution
Lamiales / Lamiaceae	*Mentha sp.*	Mint	97	
Lamiales / Lamiaceae	*Ocimum tenuiflorum*	Tulsi	52	
Lamiales / Lamiaceae	*Plectranthus amboinicus*	Mexican Mint	48	
Lamiales / Oleaceae	*Jasminum sp.*	Jasmine	71	
Lamiales / Oleaceae	*Nyctanthes arbor-tristis*	Parijata	40	
Malvales / Malvaceae	*Hibiscus rosa-sinensis*		43	
Malvales / Muntingiaceae	*Muntingia calabura*	Jamaica Cherry-Gasagase	56	
Myrtales / Lythraceae	*Punica granatum*	Pomegranate	79	
Myrtales / Myrtaceae	*Psidium guajava*	Guava	65	
Myrtales / Myrtaceae	*Syzygium cumini*	Jamun	39	
Myrtales / Myrtaceae	*Syzygium jambos*	Rose Apple	56	
Piperales / Piperaceae	*Piper betle*	Betel	48	
Rosales / Moraceae	*Artocarpus heterophyllus*	Jackfruit	56	
Rosales / Moraceae	*Ficus auriculata*	Roxburgh fig	50	
Rosales / Moraceae	*Ficus religiosa*	Peepal Tree	63	
Santalales / Santalaceae	*Santalum album*	Sandalwood	58	
Sapindales / Anacardiaceae	*Mangifera indica*	Mango	62	
Sapindales / Meliaceae	*Azadirachta indica*	Neem	60	
Sapindales / Rutaceae	*Citrus limon*	Lemon	57	
Sapindales / Rutaceae	*Murraya koenigii*	Curry	60	
Zingiberales / Zingiberaceae	*Alpinia galanga*	Rasna	50	